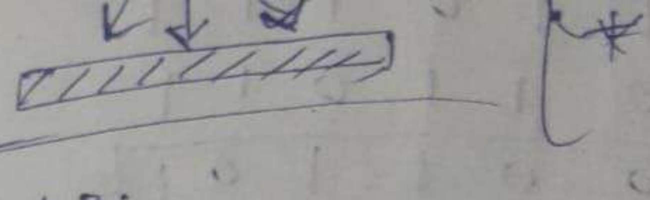
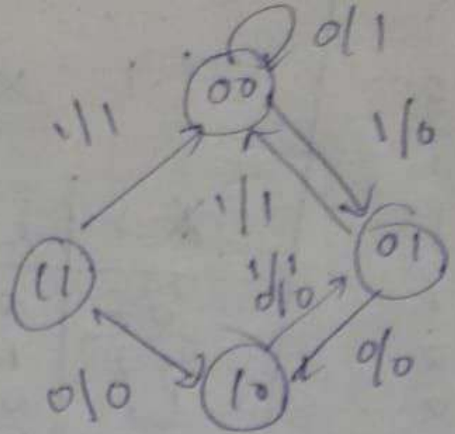


Tutorial 8 :-

#(1) All reset => & up counter

MSB		LSB	
Q_2	Q_1	Q_0	
0	0	0 ✓	=
1	0	0 ✓	= 1
0	1	0 ✓	= 0
1	1	0 ✓	=
0	0	1 ✓	
1	0	1	
$\bar{Q}_2$		$\bar{Q}_0$	
0		0	

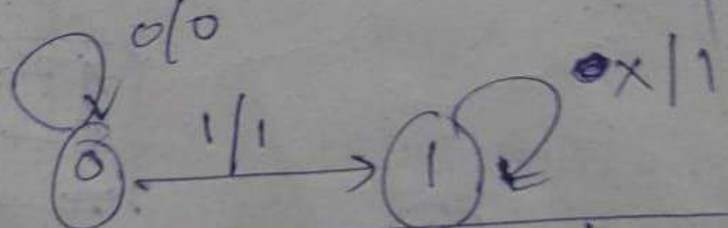
0 1 → 1
1 1 → 1
0 1 → 1
1 0 → 1
0 0 => 0 $\boxed{\text{OR}}$ gate ✓



#(5) a → 00
b → 01
c → 10

	Q_1	Q_0	y	Q_1^+	Q_0^+	O/P
a	0	0	0	0	0	0
	0	0	1	1	0	1
b	0	1	0	1	0	0
	0	1	1	0	0	0
c	1	0	0	1	0	1
	1	0	1	0	0	1
	1	1	0	x	x	x
	1	1	1	x	x	x

#(4)



	i/P , N/S		O/P
a	0, a	1, b	0, 1
b	0, b	1, b	1

1

Extra clues :-

#ch	1	1	0	1	0	0	1	1	0	1	0	1	1	0
	0	0	0	1	0	0	0	0	0	1	0	1	0	0

$\boxed{A_1 > A_2} \Rightarrow \boxed{X = 1 ; Y = 0}$

$\boxed{A_1 < A_2} \Rightarrow \boxed{X = 0 ; Y = 1}$

Both

#7(c) Initially $\rightarrow (1111)_2$

A_3	A_2	A_1	A_0	
1	1	1	1	= 15
1	1	1	0	=
1	1	0	1	=
1	1	0	0	= 12
1	0	1	1	= 11
1	0	1	0	
1	0	0	1	
1				

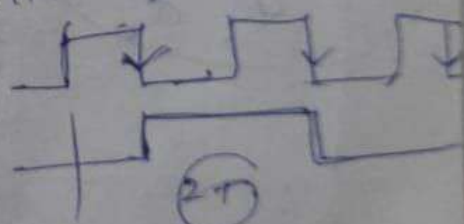
$\boxed{Y = I_7, I_6, 0, 0, I_5, I_4, 0, 0 \ldots}$

#8

Q_3	Q_2	Q_1	Q_0
f/8		f/4	f/2
0	0	0	0
0	0	0	1
0	0	1	0
0	0	1	1
0	1	0	0
0	1	0	1
0	1	1	0
0	1	1	1
1	0	0	0
1	0	0	1
1	0	1	0
1	0	1	1
1	1	0	0
1	1	0	1
1	1	1	0
1	1	1	1

f

1 Hz



2T

$Q_0 \rightarrow 2T$

Q_{n+1}	$\bar{T}\bar{Q}$	$\bar{T}Q$	TQ	$T\bar{Q}$
$\bar{S}\bar{R}$	0	1	0	1
$\bar{S}R$	0	0	0	1
SR	0	1	0	1
$S\bar{R}$	1	1	0	1

$S\bar{R}\bar{T}$

$\therefore \quad T\bar{Q} + S\bar{R}\bar{Q} + S\bar{T}Q + \bar{R}\bar{T}Q$

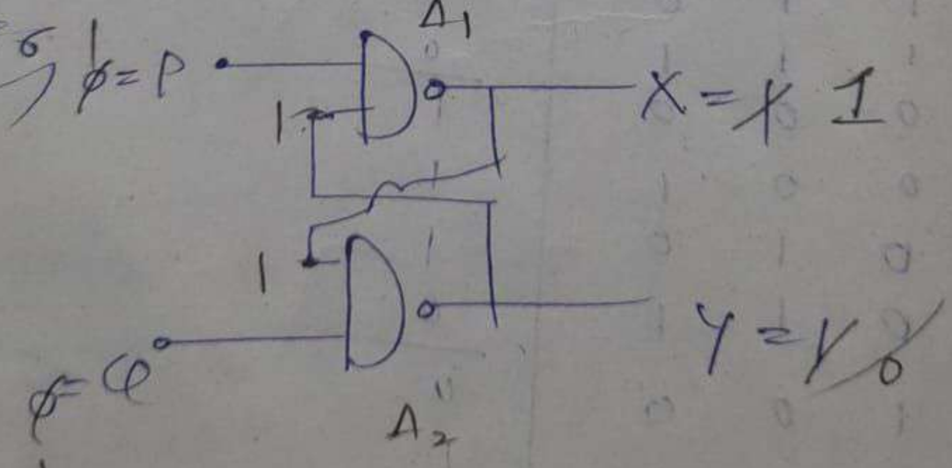
$\bar{T}Q($

$$T\bar{Q} + \bar{T}(S\bar{R} + SQ + Q\bar{R})$$

$$= \boxed{T\bar{Q} + \bar{T}(Q(S+\bar{R}) + S\bar{R})}$$

$$\boxed{T\bar{Q} + \bar{T}[Q(S+\bar{R}) + S\bar{R}]} \checkmark$$

Eg 6



#(2)

x_7	x_6	x_5	x_4	x_3	x_2	x_1	x_0
0	1	1	0	1	1	1	0
0	0	1	1	0	1	1	1
1	0	0	1	1	0	1	1
0	1	0	0	1	1	0	1

#(3)

e_1 e_2 e_3 e_4
1 0 1 0

1 0 1 0
1 1 0 1
0 1 1 0
1 0 1 1
0 1 0 1
0 0 1 0
1 0 0 1
0 1 0 0
1 0 1 0

$0 \rightarrow 1$
$1 \rightarrow 0$

8

#4)

	S	Co	A	B	Cin
	0	0	0	0	0
↳	0	1	1	1	0
↳	1	1	1	1	1

Cin = 1
A = 1
B = 1

$\boxed{S = 1 ; C_0 = 1,}$

#5)

S	R	T	Q	Qn+1
X	X	1		
1	0	0	1	0

X

	S	R	T	Qn	Qn+1
000	0	0	0	0	0
	0	0	0	1	1
T=1	0	0	1	0	1
	0	0	1	1	0
010	0	1	0	0	0
	0	1	0	1	0
T=1	0	1	1	0	1
	0	1	1	1	0
100	1	0	0	0	1
	1	0	0	1	1
T=1	1	0	1	0	1
	1	0	1	1	0
110	1	1	0	0	0
	1	1	0	1	1
T=1	1	1	1	0	1
	1	1	1	1	0

a_3	a_2	a_1	a_0
0	0	0	0
1	0	0	0
1	1	0	0
1	1	1	0
1	1	1	1
0	1	1	1
0	0	1	1
0	0	0	1
0	0	0	0

4-bit

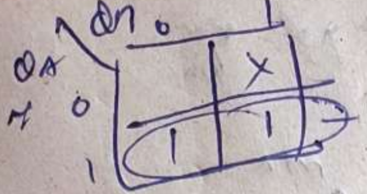
$\boxed{8}$

$\left(\frac{f_i}{8}\right)$

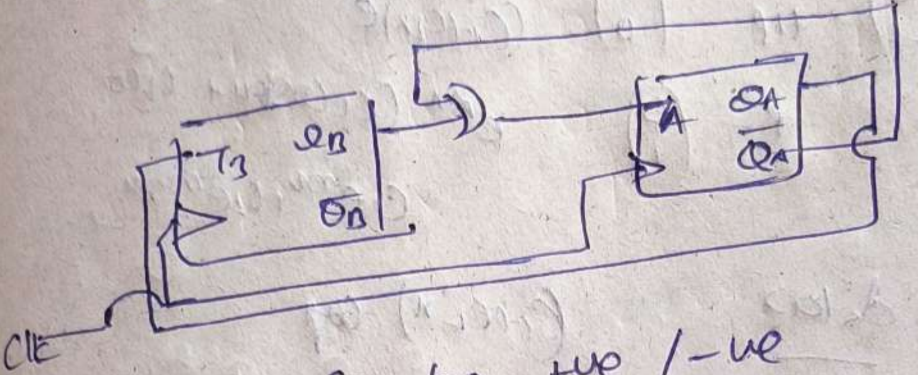
$\therefore \left(\frac{f_i}{64}\right)$

$= \left(\frac{3.6\,KHz}{64}\right)$

$= \boxed{56.25\,(Hz)}$



$T_B = Q_A$

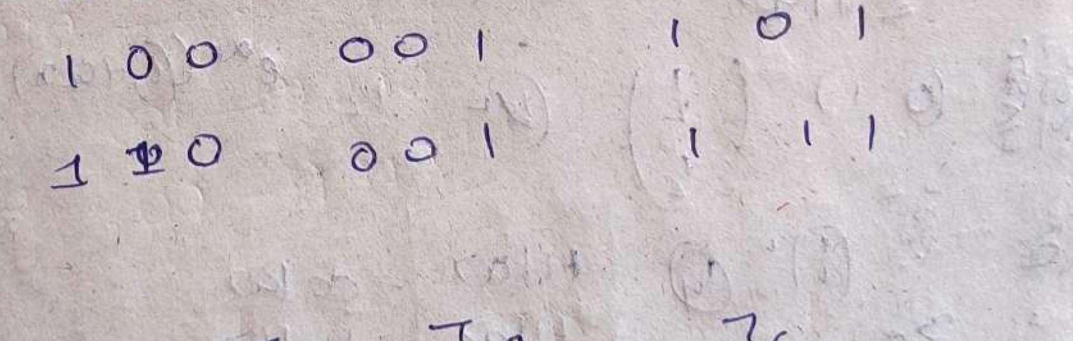


Clk can be +ve / -ve

	T_A	T_B	T_C
14			

Q_A	Q_B	Q_C	next	T_A	T_B	T_C
0	0	1	0 1 0	0	1	1
0	1	0	0 1 1	0	0	1
0	1	1	1 0 1	1	1	0
1	0	1	1 1 1	0	1	0
1	1	1	0 0 1	1	1	0
0	0	0	0 0 1	0	0	1
1	0	0	0 0 1	1	0	1
1	1	0	0 0 1	1	1	1

solve for T_A T_B T_C



T_A

(12) fig(b)

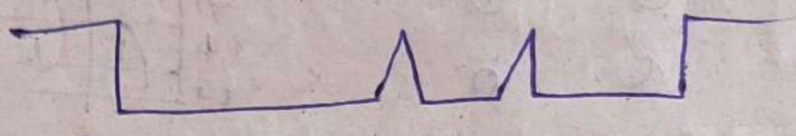
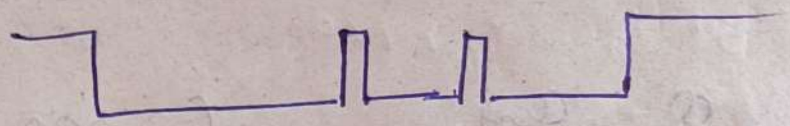
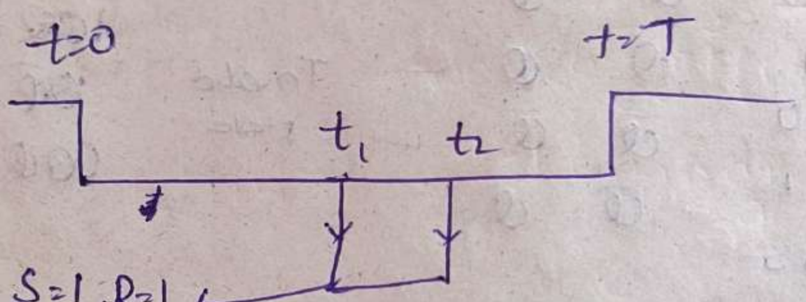


fig b





@ $S=1, R=1$ ←
t_1, t_2 ⇒ Hold

remaining time $\left.\begin{matrix} S=1 \\ R=0 \end{matrix}\right\}$ in ~~first~~ given ckt

O/P of R becomes '0'

∴ overall o/p = 0

(13) $\overline{Q_2 \cdot Q_0} = 0$ ⇒

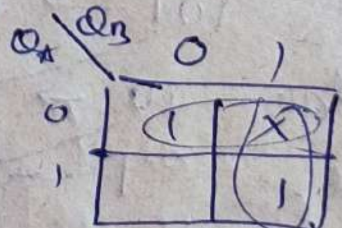
Q_2	Q_1	Q_0
0	0	0
⋮		
1	0	1

} total 5 States

∴ MOD 5

(14a)

Pres ($Q_A Q_B$)	Next ($Q_A^+ Q_B^+$)	T_A	T_B
00	10	1	0
10	11	0	1
11	00	1	1
01	X X	X	X



$T_A = \overline{Q_A} + Q_B$

(9a)

X	Y	Q	Q_{n+1}
0	0	X	0
0	1	Q	Q
1	0	Q	$\bar{Q}$
1	1	X	1

XY \ Q	0	1
00		
01		1
11	1	1
10	1	

$Q_{n+1} = \boxed{X\bar{Q} + YQ}$

(9b)

X	Y	Q	Q_{n+1}
0	0	Q	$\bar{Q}$
0	1	Q	Q — Invalid state
1	0	$\bar{Q}$	$\bar{Q}$ —
1	1	$\bar{Q}$	Q

$(\because 00\bar{Q} = Q$
$10Q = Q$
$10\bar{Q} = \bar{Q}$
$00Q = \bar{Q})$

0000

(10)

~~1100~~ → 0000

i.e. 12 states ∴ MOD-12

$(\because \overline{Q_2 \cdot Q_3} = \overline{1 \cdot 1} = 0 = CLR)$

(11a)

0010 → $\boxed{0100}$

→ 1st 001 → 0000

→ 2nd clk 00 → $\boxed{1000}$

(11b)

1011 → $\boxed{1101}$

1st clk 101 → 1110

2nd clk 10 → 1111

3rd clk 1 → $\boxed{0111}$